StudyCircle — Software Requirements Specification

Oct 15, 2026 · @Pooja Joshi

1. Introduction

StudyCircle is a MERN web app where students form private, admin-approved study groups to share notes, chat with @mentions, video call and get AI help. This SRS defines every feature and rule the team must build and test against.

1.1 Purpose

This document is the single reference for what StudyCircle must do and how well it must do it. Developers build from it, testers write test cases from its requirement IDs, and reviewers use it to judge scope.

1.2 Scope

In scope:

- Account sign-up and login, with a mobile number visible only to the platform superadmin
- Study groups with admin approval, roles, and membership in many groups per user
- Shared group notes (files and Markdown), search, upvotes and bookmarks
- Real-time group chat with @mentions, @everyone and notifications
- Peer-to-peer video calls for up to 5 people per group
- Public group reviews that only the superadmin can remove
- Shared Pomodoro timer, AI study features and a superadmin panel

Out of scope for this release: native mobile apps, payments, one-to-one direct messages, calls larger than 5 people, and SMS OTP verification.

1.3 Definitions

Term	Meaning
Superadmin	Platform owner role; manages all users, groups and reviews
Group admin	Creator or owner of one group; manages that group only
Moderator (mod)	Member promoted by a group admin to help manage the group
Member	Approved user inside a group
Join request	A user's request to enter an approval-only group
Mention	A reference to a member in chat with `@`, which notifies them
StudyBot	The AI assistant account that answers when mentioned
Mesh	WebRTC layout where every call participant connects to every other
RAG	Retrieval-augmented generation: AI answers grounded in the group's own notes
SFU	Selective forwarding unit, a media server for larger calls

1.4 References

- Earlier project context notes, shared in the project chat and superseded by this SRS where they differ (`PROJECT_CONTEXT.md`)
- StudyCircle UI design canvas (7 screens)
- IEEE 830 structure for software requirements specifications

1.5 Requirement notation

Each requirement has an ID such as **FR-AUTH-03** and a priority: **Must** (V1, required to ship), **Should** (V2) or **Could** (V3). "The system shall" states an obligation that must be testable.

2. Overall description

StudyCircle is a standalone browser app: a React client talks to one Node.js/Express server over REST and Socket.io, backed by MongoDB Atlas, Cloudinary and an AI provider.

2.1 Product perspective

The product replaces the mix of WhatsApp groups, Drive folders and separate video links that study groups use today. It is a new system with no legacy data to migrate. Video travels browser-to-browser (WebRTC); the server only relays connection setup messages.

2.2 Product functions

1. Register, log in and manage a private profile
2. Discover, create and join study groups
3. Approve or reject join requests and manage member roles
4. Share, view, search and rate notes inside a group
5. Chat in real time with @mentions and notifications
6. Start and join group video calls with screen share
7. Read and write public group reviews
8. Study together on a shared Pomodoro timer
9. Use AI to summarise notes and answer doubts in chat
10. Moderate the whole platform as superadmin

2.3 User classes

User class	Who	What they can do	Expected skill
Visitor	Not logged in	Browse public group info and reviews, sign up	Basic web use
Member	Approved user in a group	Notes, chat, calls, reviews, timer, AI	Basic web use
Moderator	Member promoted by admin	Member rights plus approve requests, delete messages and notes, control timer and calls	Basic web use

User class	Who	What they can do	Expected skill
Group admin	Group creator or owner	Moderator rights plus roles, removals, settings, one reply per review	Basic web use
Superadmin	Platform owner	All users and groups, mobile numbers, review removal, bans, audit log	Technical

One person can hold different group roles in different groups at the same time. Platform role (user or superadmin) and group role are independent.

2.4 Operating environment

- Client: latest two versions of Chrome, Edge, Firefox and Safari on desktop and mobile; HTTPS required for camera and microphone
- Server: Node.js LTS with Express and Socket.io on a host that supports WebSockets (Render or Railway)
- Database: MongoDB Atlas; file storage: Cloudinary; frontend hosting: Vercel

2.5 Design and implementation constraints

- Stack is fixed: MongoDB, Express, React (Vite), Node.js, Tailwind CSS, Socket.io
- Video uses a WebRTC mesh, so one call is capped at 5 participants
- AI provider keys live only on the server; the browser never calls the AI API directly
- Free or low-cost tiers of hosting, storage and AI services; usage must stay within their limits
- Personal data handling follows India's DPDP Act principles: purpose, consent and minimum exposure

2.6 Assumptions and dependencies

- Users have a stable internet connection and a webcam for calls
- Google's public STUN server is reachable; calls on strict networks may fail without a TURN server
- Cloudinary, MongoDB Atlas and the AI provider stay available within their free or paid tiers

- Most users are in India, so time is shown in IST by default

3. Functional requirements

The system has 12 modules and 125 numbered requirements; every one is enforced on the server, whatever the UI shows.

3.1 Authentication and accounts (AUTH)

ID	Requirement	Priority
FR-AUTH-01	The system shall let a visitor register with full name, Gmail address, Indian mobile number and password.	Must
FR-AUTH-02	The system shall accept only mobile numbers of 10 digits starting with 6, 7, 8 or 9, stored with the +91 prefix.	Must
FR-AUTH-03	The system shall reject registration when the email or mobile number already belongs to an account.	Must
FR-AUTH-04	The system shall require passwords of at least 8 characters and store them only as bcrypt hashes.	Must
FR-AUTH-05	The system shall show this consent line on sign-up: mobile number used only for account recovery and visible only to platform administrators.	Must
FR-AUTH-06	The system shall log users in with email and password and issue a JWT holding only user ID and platform role.	Must
FR-AUTH-07	The system shall expire tokens after 7 days and send the user to login when a token is invalid or expired.	Must
FR-AUTH-08	The system shall let a logged-in user view and edit their own name and avatar, and change their mobile number only after re-entering their password (SMS OTP replaces this check once FR-AUTH-11 ships).	Must

ID	Requirement	Priority
FR-AUTH-09	The system shall block login for banned accounts and show the reason.	Should
FR-AUTH-10	The system shall offer "Continue with Google" sign-in, then ask for a mobile number on first login.	Could
FR-AUTH-11	The system shall verify the mobile number by SMS OTP.	Could
FR-AUTH-12	The system shall let a user delete their account, removing all memberships and handing over or deleting groups they own.	Should

3.2 Privacy of personal data (PRIV)

ID	Requirement	Priority
FR-PRIV-01	The system shall never show a user's mobile number to anyone except that user and the superadmin.	Must
FR-PRIV-02	The system shall never show a user's email to other users, group admins or moderators.	Must
FR-PRIV-03	The system shall expose other users only as name and avatar in every API response, member list and real-time event.	Must
FR-PRIV-04	The system shall exclude email, mobile and password hash from database queries by default (`select: false`).	Must
FR-PRIV-05	The system shall return mobile numbers only from superadmin routes behind a superadmin check.	Must
FR-PRIV-06	The system shall mask mobile numbers in the superadmin list (last 4 digits) until the superadmin clicks Reveal.	Should
FR-PRIV-07	The system shall write an audit log entry each time a superadmin reveals a mobile number.	Should

ID	Requirement	Priority
FR-PRIV-08	The system shall encrypt mobile numbers at rest and keep a separate hash for uniqueness checks.	Could

3.3 Groups and membership (GRP)

ID	Requirement	Priority
FR-GRP-01	The system shall let any logged-in user create a group with name, description, subject tags and privacy mode.	Must
FR-GRP-02	The system shall make the creator the group admin.	Must
FR-GRP-03	The system shall support three privacy modes: Open (join instantly), Approval (default, admin approves) and Invite only (hidden from search).	Must
FR-GRP-04	The system shall generate a unique 6-character invite code per group and let users join with it.	Must
FR-GRP-05	The system shall let one user belong to many groups at once.	Must
FR-GRP-06	The system shall cap membership at 20 groups per user and creation at 5 groups per user (configurable).	Should
FR-GRP-07	The system shall list the user′s groups in a sidebar with unread count and mention badge per group.	Must
FR-GRP-08	The system shall let users search Open and Approval groups by name, tag or exam, and filter by exam chips.	Must
FR-GRP-09	The system shall show a public preview page with description, member count, notes count, privacy mode, rating and reviews.	Must
FR-GRP-10	The system shall let a group admin edit the group′s name, description, tags, privacy and settings.	Must

ID	Requirement	Priority
FR-GRP-11	The system shall let a group admin promote a member to moderator or demote a moderator.	Must
FR-GRP-12	The system shall let a group admin remove a member, immediately ending their access to chat, notes and any active call.	Must
FR-GRP-13	The system shall let any member leave a group.	Must
FR-GRP-14	The system shall stop a group admin leaving until they transfer admin to another member.	Must
FR-GRP-15	The system shall let a group admin delete the group after a typed confirmation.	Must
FR-GRP-16	The system shall let a user mute notifications for one group.	Should
FR-GRP-17	The system shall hide Invite-only groups from search and give them no public preview; their preview page opens only through a valid invite link.	Must

3.4 Join requests (REQ)

ID	Requirement	Priority
FR-REQ-01	The system shall let a user send a join request to an Approval group, with an optional message of up to 300 characters.	Must
FR-REQ-02	The system shall allow only one pending request per user per group.	Must
FR-REQ-03	The system shall show the user's request state on the group card: Request to join, Pending, or Joined.	Must
FR-REQ-04	The system shall list pending requests to the group's admin and moderators with requester name, time and message.	Must
FR-REQ-05	The system shall let the admin or a moderator approve or reject each request.	Must

ID	Requirement	Priority
FR-REQ-06	On approval, the system shall create the membership and notify the requester in real time.	Must
FR-REQ-07	On rejection, the system shall notify the requester and allow a new request after 7 days.	Must
FR-REQ-08	The system shall notify the admin and moderators in real time when a new request arrives.	Must
FR-REQ-09	The system shall let a user cancel their own pending request.	Should

3.5 Group notes (NOTE)

ID	Requirement	Priority
FR-NOTE-01	The system shall let members upload PDF, PNG, JPG and DOCX files up to 10 MB to a group.	Must
FR-NOTE-02	The system shall check file type and size on the server, not only in the browser.	Must
FR-NOTE-03	The system shall let members write notes in Markdown inside the app.	Must
FR-NOTE-04	The system shall store each note with title, folder or chapter, tags, uploader and date.	Must
FR-NOTE-05	The system shall show PDFs and images in an in-app viewer and allow download.	Must
FR-NOTE-06	The system shall let members search notes by title and tag and filter by folder.	Must
FR-NOTE-07	The system shall let only the uploader edit a note, and only the uploader, a moderator or the admin delete it.	Must
FR-NOTE-08	The system shall extract and store the text of uploaded PDFs for search and AI use.	Should
FR-NOTE-09	The system shall let members mark a note "helpful" once and sort notes by helpful count.	Should

ID	Requirement	Priority
FR-NOTE-10	The system shall let members bookmark notes into a personal "Saved" list.	Should
FR-NOTE-11	The system shall notify members when a new note is added.	Should
FR-NOTE-12	The system shall let members comment on a note.	Could

3.6 Group chat and mentions (CHAT)

ID	Requirement	Priority
FR-CHAT-01	The system shall give each group one real-time chat visible only to its members.	Must
FR-CHAT-02	The system shall deliver a sent message to all online members within 1 second on a normal connection.	Must
FR-CHAT-03	The system shall save every message and load history 30 at a time as the user scrolls up.	Must
FR-CHAT-04	The system shall show the sender's message instantly and mark it failed with a retry button if the server rejects it.	Must
FR-CHAT-05	The system shall open a member picker when the user types `@` and insert the chosen member.	Must
FR-CHAT-06	The system shall send mentioned user IDs with the message and keep only IDs that belong to current group members.	Must
FR-CHAT-07	The system shall highlight messages that mention the reader.	Must
FR-CHAT-08	The system shall notify a mentioned user even when they are on another page or group.	Must
FR-CHAT-09	The system shall open the group at the mentioned message and highlight it when the user clicks a mention notification.	Must

ID	Requirement	Priority
FR-CHAT-10	The system shall allow `@everyone` only for the admin and moderators.	Must
FR-CHAT-11	The system shall show a "typing…" indicator for other members.	Must
FR-CHAT-12	The system shall let the admin and moderators delete any message, shown as "Message removed".	Must
FR-CHAT-13	The system shall limit messages to 2,000 characters.	Must
FR-CHAT-14	The system shall let a member attach a group note to a message.	Should
FR-CHAT-15	The system shall let a sender edit their message within 15 minutes, marked "Edited".	Should
FR-CHAT-16	The system shall support replies to a specific message and emoji reactions.	Could
FR-CHAT-17	The system shall let the admin and moderators pin up to 3 messages.	Could

3.7 Notifications (NOTIF)

ID	Requirement	Priority
FR-NOTIF-01	The system shall create notifications for join requests, approvals, rejections, mentions, new notes, call starts and review replies.	Must
FR-NOTIF-02	The system shall deliver notifications live to every open tab of the user and store them for later.	Must
FR-NOTIF-03	The system shall show unread notifications in a bell with a count, and let the user mark them read.	Must
FR-NOTIF-04	The system shall not send new-message notifications from muted groups, except mentions.	Should

3.8 Video calls (CALL)

ID	Requirement	Priority
FR-CALL-01	The system shall let any member start or join the group's video call.	Should
FR-CALL-02	The system shall check group membership on the server before a user joins a call.	Should
FR-CALL-03	The system shall cap a call at 5 participants and show "Call is full" to the 6th.	Should
FR-CALL-04	The system shall show a live call indicator with participant count in the group header and sidebar.	Should
FR-CALL-05	The system shall let participants mute and unmute the microphone and turn the camera on and off.	Should
FR-CALL-06	The system shall let a participant share their screen and label their tile "sharing screen" for everyone in the call.	Should
FR-CALL-07	The system shall highlight the participant who is speaking with a ring on their tile and a "Speaking" label in the participant list.	Should
FR-CALL-08	The system shall let the admin or a moderator mute a participant or end the call for everyone.	Should
FR-CALL-09	The system shall allow a user in only one call at a time; joining another leaves the first.	Should
FR-CALL-10	The system shall release the camera and microphone when the user leaves or closes the tab.	Should

3.9 Group reviews (REV)

ID	Requirement	Priority
FR-REV-01	The system shall let an eligible member write one review per group: 1 to 5 stars and a comment up to 1,000 characters.	Should

ID	Requirement	Priority
FR-REV-02	The system shall make a member eligible only after 3 days in the group, and never the group's creator or its current admin (including an admin who received the group by transfer).	Should
FR-REV-03	The system shall show reviews publicly, including to visitors who are not logged in.	Should
FR-REV-04	The system shall let the author edit their review and label it "Edited".	Should
FR-REV-05	The system shall not let the group admin, moderators or members delete any review.	Should
FR-REV-06	The system shall let only the superadmin remove a review, with a written reason, as a soft delete.	Should
FR-REV-07	The system shall keep a review after its author leaves or is removed from the group.	Should
FR-REV-08	The system shall let the group admin post one public reply per review.	Should
FR-REV-09	The system shall let any logged-in user report a review with a reason.	Should
FR-REV-10	The system shall show the average rating, review count and a 5-to-1 star breakdown, updated whenever reviews change.	Should
FR-REV-11	The system shall let the superadmin remove a review only for abuse, spam, personal data or off-topic content; a negative opinion alone is not grounds for removal, and the moderation queue shall state this rule.	Should
FR-REV-12	The system shall show on each review how long the author has been (or was) a member, such as "Member for 3 months", and "left the group" when they are no longer a member.	Should

3.10 Shared Pomodoro timer (TIMER)

ID	Requirement	Priority
FR-TIMER-01	The system shall give each group one shared timer with focus (default 25 min) and break (default 5 min) modes.	Should
FR-TIMER-02	The system shall let the admin and moderators start, pause and reset the timer.	Should
FR-TIMER-03	The system shall show every member the same remaining time, within 1 second, including members who join mid-session.	Should
FR-TIMER-04	The system shall switch focus and break automatically and record a completed session for each member present.	Should
FR-TIMER-05	When the group setting is on, the system shall make chat read-only during focus mode.	Should
FR-TIMER-06	The system shall track daily study streaks and a weekly leaderboard per group.	Could

3.11 AI features (AI)

ID	Requirement	Priority
FR-AI-01	The system shall summarise a note into 5 bullet points when a member clicks Summarize.	Should
FR-AI-02	The system shall save each summary and reuse it instead of calling the AI again.	Should
FR-AI-03	The system shall reply as StudyBot in the group chat when a member mentions @StudyBot, using the last 10 messages as context.	Should
FR-AI-04	The system shall let the group admin turn StudyBot on or off for the group.	Should

ID	Requirement	Priority
FR-AI-05	The system shall limit each user to 20 AI requests per day and show a clear message when the limit is reached.	Should
FR-AI-06	The system shall label all AI-generated content with an AI marker.	Should
FR-AI-07	The system shall show a friendly retry message when the AI service fails or times out after 30 seconds.	Should
FR-AI-08	The system shall generate a 5–10 question quiz from a note.	Could
FR-AI-09	The system shall summarise a user's unread messages in a group on request ("Catch me up").	Could
FR-AI-10	The system shall answer questions using only the group's notes and cite which note each answer came from (RAG).	Could

3.12 Superadmin panel (ADM)

ID	Requirement	Priority
FR-ADM-01	The system shall restrict the admin panel to accounts with the superadmin platform role.	Should
FR-ADM-02	The system shall show platform totals: users, groups, notes and live calls.	Should
FR-ADM-03	The system shall list and search all users with name, email, masked mobile and group count.	Should
FR-ADM-04	The system shall let the superadmin ban and unban a user, ending their active sessions.	Should
FR-ADM-05	The system shall list all groups and let the superadmin delete a group.	Should

ID	Requirement	Priority
FR-ADM-06	The system shall show a queue of reported reviews with report count and reasons, and Keep or Remove actions.	Should
FR-ADM-07	The system shall record mobile reveals, review removals, bans and group deletions in an audit log with who, what and when.	Should
FR-ADM-08	The system shall let the superadmin view the audit log, newest first.	Should

4. External interface requirements

The client and server talk through a JSON REST API under `/api` and a Socket.io connection; both require a valid JWT except the public routes marked below.

4.1 User interface

- Visual style: light frosted glass on a soft off-white background, blue as the main colour, teal for AI, amber for mentions and ratings (see the UI design canvas)
- Layout: left sidebar of the user's groups; group room with tabs Chat, Notes, Members, Reviews; video call button in the group header
- Screens: Sign up and Login, Discover groups, Group preview with reviews, Group room, Video call, Manage members (admin), Superadmin panel, Profile
- Responsive from 360 px phone width to 1440 px desktop; the sidebar becomes a slide-out drawer on phones
- Motion: page and list transitions of 150–300 ms, disabled when the user's system asks for reduced motion
- Every action shows loading, success and error states; errors say what happened and what to do

The design canvas shows the full product across V1–V3, so some screens include Should and Could features (video calls, reviews, the superadmin panel, streaks, Google sign-in). Each feature ships only in the release phase its priority sets (section 7).

4.2 REST API

Area	Method and path	Access
Auth	`POST /api/auth/register`, `POST /api/auth/login`	Public
Profile	`GET /api/me`, `PATCH /api/me`	Self
Groups	`GET /api/groups?search=&tags=`	Public
Groups	`GET /api/groups/:id` (preview)	Public
Groups	`POST /api/groups`, `GET /api/groups/mine`, `POST /api/groups/join/:inviteCode`	Logged in
Groups	`PATCH /api/groups/:id`, `DELETE /api/groups/:id`	Group admin
Join requests	`POST /api/groups/:id/requests`	Logged in
Join requests	`GET /api/groups/:id/requests`, `PATCH /api/groups/:id/requests/:reqId`	Admin or mod
Join requests	`DELETE /api/groups/:id/requests/:reqId` (cancel own pending request)	Requester
Members	`GET /api/groups/:id/members`	Member
Members	`PATCH` / `DELETE /api/groups/:id/members/:userId`	Group admin
Members	`DELETE /api/groups/:id/members/me`	Member
Members	`PATCH /api/groups/:id/members/me` (mute or unmute the group)	Member
Members	`POST /api/groups/:id/transfer-admin` (to another member, FR-GRP-14)	Group admin
Notes	`GET` / `POST /api/groups/:id/notes`	Member
Notes	`PATCH` / `DELETE /api/groups/:id/notes/:noteId`	Uploader, mod, admin

Area	Method and path	Access
Notes	`POST /api/groups/:id/notes/:noteId/upvote`, `/summary`	Member
Notes	`POST` / `DELETE /api/groups/:id/notes/:noteId/bookmark`, `GET /api/me/saved`	Member
Messages	`GET /api/groups/:id/messages?before=`	Member
Reviews	`GET /api/groups/:id/reviews`	Public
Reviews	`POST` /api/groups/:id/reviews, `PATCH /api/groups/:id/reviews/:rid`	Eligible member, author
Reviews	`POST /api/groups/:id/reviews/:rid/reply`	Group admin
Reviews	`POST /api/groups/:id/reviews/:rid/report`	Logged in
Notifications	`GET /api/notifications`, `PATCH /api/notifications/read`	Self
Admin	`GET /api/admin/users`, `/groups`, `/reviews/reported`, `/stats`, `/audit-log`	Superadmin
Admin	`PATCH /api/admin/users/:id/ban`, `DELETE /api/admin/groups/:id`, `DELETE /api/admin/reviews/:rid`	Superadmin
Admin	`GET /api/admin/users/:id/mobile` (Reveal; writes an audit log entry)	Superadmin
Admin	`PATCH /api/admin/users/:id/unban`, `PATCH /api/admin/reviews/:rid/keep` (clears its reports)	Superadmin

Errors return `{ "error": { "code", "message" } }` with status 400 (invalid input), 401 (not logged in), 403 (not allowed), 404 (not found), 409 (duplicate) or 429 (rate limited).

4.3 Real-time events (Socket.io)

On connect the server checks the JWT, then joins the socket to `user:<id>` and to `group:<id>` for each membership.

Direction	Event	Purpose
Client → server	`chat:send`, `chat:typing`, `chat:read`	Send a message, show typing, mark read
Client → server	`video:join`, `video:signal`, `video:leave`	Join a call, relay WebRTC setup, leave
Client → server	`timer:start`, `timer:pause`, `timer:reset`	Control the shared timer (admin or mod)
Client → server	`chat:edit`	Edit own message within 15 minutes
Client → server	`video:mute`, `video:end`	Mute a participant or end the call for everyone (admin or mod)
Server → client	`chat:new`, `chat:deleted`, `chat:typing`	Live chat updates
Server → client	`chat:edited`	Show the new text marked "Edited"
Server → client	`mention:new`, `notification:new`	Personal alerts to `user:<id>`
Server → client	`request:new`, `request:reviewed`	Join request alerts
Server → client	`note:new`, `call:started`, `call:ended`	Group activity
Server → client	`video:user-joined`, `video:user-left`, `video:signal`	WebRTC connection setup
Server → client	`video:muted`, `video:ended`	A mod muted you, or the call was ended for everyone
Server → client	`timer:state`	Timer mode, start time, duration, running
Server → client	`member:removed`, `member:role-changed`	On removal the server also takes the user's sockets out of `group:<id>` and drops them from any active call (FR-GRP-12)

User objects inside any event carry only `_id`, `name` and `avatar`.

4.4 Third-party services

Service	Used for	Interface
MongoDB Atlas	All application data, text and vector search	Mongoose driver
Cloudinary	Note files and avatars	Server-side upload API; only URLs stored
AI provider (Claude, Gemini or OpenAI)	Summaries, StudyBot, quizzes, RAG	Server-side SDK; key in `.env`
Google STUN	WebRTC network discovery	`stun:stun.l.google.com:19302`
Google OAuth	Optional sign-in	OAuth 2.0 (Could)

5. Data requirements

The system stores 11 MongoDB collections; users and groups are linked many-to-many through a separate Membership collection.

5.1 Entities

Collection	Key fields	Indexes and rules
User	name, avatar, email*, mobile*, passwordHash*, platformRole (user, superadmin), isBanned, streak	Unique email, unique mobile; * hidden by default (`select: false`)
Group	name, description, tags, privacy (open, approval, invite), inviteCode, createdBy, settings (studyBotEnabled, chatLockDuringFocus), timer, activeCall, ratingAverage, ratingCount	Unique inviteCode; text index on name and tags

Collection	Key fields	Indexes and rules
Membership	user, group, role (admin, mod, member), joinedAt, lastReadAt, isMuted	Unique (user, group); index on group
JoinRequest	group, user, message, status (pending, approved, rejected), reviewedBy, createdAt, reviewedAt	Partial unique (group, user) where status is pending, so re-requests after a rejection never collide; index (group, status)
Message	group, sender, text, mentions, replyTo, attachedNote, isEdited, isDeleted, createdAt	Index (group, createdAt desc)
Note	group, uploadedBy, title, type (file, markdown), fileUrl, fileType, content, extractedText, folder, tags, upvotes, bookmarkedBy, summary	Text index on title and tags
Review	group, author, rating (1–5), comment, isEdited, adminReply, reports, isRemoved, removedBy, removedReason	Unique (group, author)
Notification	user, type, group, refId, read, createdAt	Index (user, read, createdAt)
AuditLog	action, by, target, meta, createdAt	Index on createdAt; append-only
StudySession	user, group, minutes, completedAt	Index (group, completedAt)
NoteChunk	note, group, text, embedding	Vector index on embedding (V3)
StudyBot account	A User record with a reserved ID, used as the sender of AI replies	Cannot log in

5.2 Data rules

- Messages, memberships and chunks are separate collections, never arrays inside Group, so no document grows without limit (MongoDB caps documents at 16 MB).
- Reviews, messages and removed reviews are soft-deleted (`isDeleted` / `isRemoved`) to keep an audit trail.

- Group rating average and count are stored on the Group and recalculated on every review change.
- All timestamps are stored in UTC and shown in the user's time zone (IST by default).

5.3 Retention

Data	Kept for
Account data	Until the user deletes their account
Messages and notes	Until the group is deleted, or the item is deleted by an allowed role
Reviews	Until the superadmin removes them; kept after the author leaves
Notifications	90 days
Audit log	1 year
Deleted account	Personal fields erased within 30 days; their reviews show "Former member"

6. Non-functional requirements

The app must be secure by default, keep personal data private, and feel instant on an ordinary Indian 4G or broadband connection.

6.1 Security

ID	Requirement
NFR-SEC-01	Every permission shall be checked on the server, for both REST routes and socket events; hidden buttons are never the protection.
NFR-SEC-02	All traffic shall use HTTPS and WSS.
NFR-SEC-03	Passwords shall be hashed with bcrypt (cost 10 or higher).

ID	Requirement
NFR-SEC-04	Login and register shall be limited to 10 attempts per 15 minutes per IP; AI routes to 20 requests per user per day.
NFR-SEC-05	All inputs shall be validated on the server (types, lengths, allowed values) and chat and Markdown rendered without raw HTML.
NFR-SEC-06	CORS shall allow only the frontend origin; security headers shall be set with helmet.
NFR-SEC-07	Secrets (database URL, JWT secret, Cloudinary and AI keys) shall live only in server environment variables, never in the repository or client.
NFR-SEC-08	Uploaded files shall be checked for type and size on the server and served from Cloudinary, never executed.

6.2 Privacy

ID	Requirement
NFR-PRIV-01	Personal data shall be collected only for a stated purpose: email for login, mobile for account recovery.
NFR-PRIV-02	No API response, socket event, log line or error message shall contain another user's email or mobile number.
NFR-PRIV-03	Sensitive superadmin actions shall be logged with who, what and when.
NFR-PRIV-04	A user shall be able to delete their account and personal data.

6.3 Performance

ID	Requirement
NFR-PERF-01	95% of API requests shall respond within 500 ms, excluding file uploads and AI calls.
NFR-PERF-02	Chat messages shall reach online members within 1 second.
NFR-PERF-03	The first screen shall load within 3 seconds on a 4G connection.
NFR-PERF-04	The system shall support 200 concurrent users and 50 active groups on one server instance.
NFR-PERF-05	AI responses shall start streaming within 5 seconds or show a progress state.
NFR-PERF-06	Animations shall use only transform and opacity and run at 60 frames per second on mid-range phones.

6.4 Reliability and availability

ID	Requirement
NFR-REL-01	The deployed app shall target 99% monthly uptime.
NFR-REL-02	The socket client shall reconnect automatically and reload missed messages after a dropped connection.
NFR-REL-03	A failure in an outside service (AI, Cloudinary) shall not break chat, notes or calls.
NFR-REL-04	The database shall be backed up daily using the hosting provider's backups.

6.5 Usability and accessibility

ID	Requirement
NFR-USE-01	A new user shall be able to sign up and send a join request within 2 minutes without help.
NFR-USE-02	Text shall meet WCAG 2.1 AA contrast (4.5:1 for body text).
NFR-USE-03	All actions shall work by keyboard, with visible focus and labels on icon-only buttons.
NFR-USE-04	Animations shall turn off when the user's system asks for reduced motion.
NFR-USE-05	Touch targets shall be at least 44 × 44 px.
NFR-USE-06	Every screen shall work from 360 px to 1440 px wide.

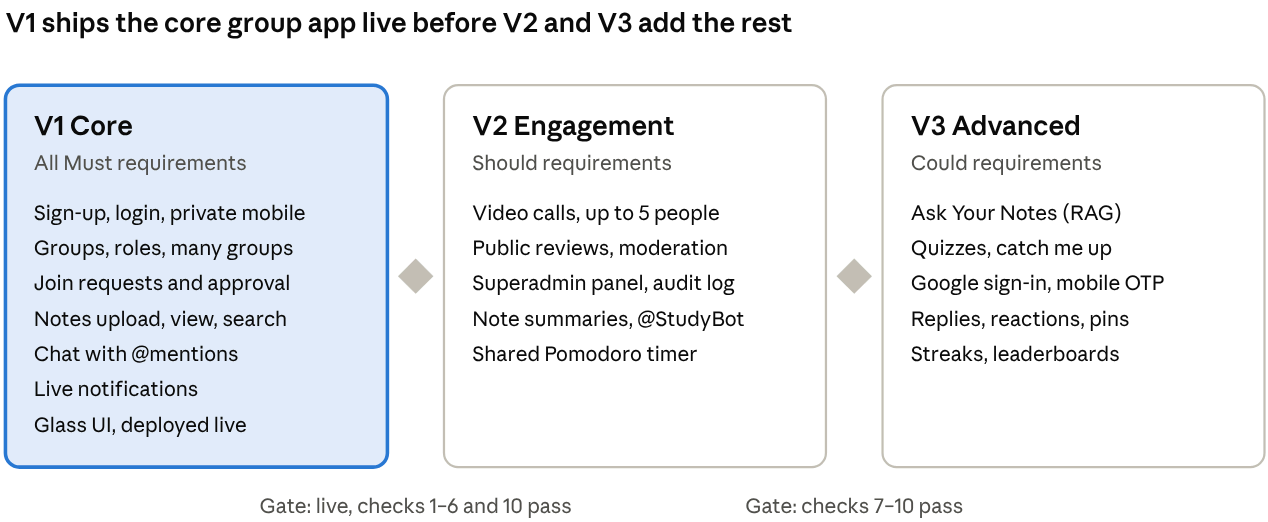
6.6 Maintainability

ID	Requirement
NFR-MNT-01	Code shall follow the agreed folder structure: models, routes, controllers, middleware, services, sockets on the server; pages, components, hooks, context on the client.
NFR-MNT-02	All AI calls shall go through one `aiService` module so the provider can be changed in one place.
NFR-MNT-03	Role checks shall use shared middleware (`auth` , `requireGroupRole` , `requireSuperadmin`), not repeated code.
NFR-MNT-04	The README shall explain setup, environment variables, architecture and known limits (mesh calls, no TURN server).

7. Release plan and acceptance

V1 ships the core group experience (all **Must** requirements) and is deployed live before any V2 work starts; V2 and V3 add the **Should** and **Could** items.

7.1 Release phases



release roadmap · 3 phases, 2 gates

Each phase starts only after the previous one passes its gate in 7.2.

7.2 Acceptance criteria

A release is accepted when every requirement in its scope passes these checks:

1. **Role checks hold.** A member cannot approve requests, change roles or delete others' notes; a group admin cannot delete reviews or see emails or mobiles; each blocked call returns 403.
2. **Privacy holds.** Inspecting every API response and socket payload as a normal user shows no email or mobile number of anyone else.
3. **Join flow works.** A request to an Approval group appears to the admin live; approving it lets the user into chat and notes without a page reload; rejecting it blocks a new request for 7 days.
4. **Many groups work.** One user in 3 groups gets messages, mentions and unread counts in all 3 at once.
5. **Mentions work.** Mentioning a member who is on another page notifies them within 1 second; clicking it opens the highlighted message.

6. **Notes work.** A 10 MB PDF uploads and opens in the viewer; an 11 MB file or a .exe file is refused with a clear message.
7. **Calls work (V2).** Four users on normal Wi-Fi hold a call with audio, video and screen share; a fifth joins; a sixth sees "Call is full".
8. **Reviews hold (V2).** No API call by a group admin, moderator or member can delete a review; the superadmin's removal is logged with a reason.
9. **Timer stays in sync (V2).** Two members on different devices see the same remaining time within 1 second, including one who joins mid-session.
10. **Quality bar (every release, checked at both gates).** Lighthouse accessibility score of 90 or more on the main screens; no console errors; the README explains setup.

7.3 Test approach

- Unit tests for middleware (auth, role checks) and services (rating recalculation, mention filtering)
- API tests for every route's allowed and forbidden roles
- Manual multi-browser tests for chat, mentions, calls and the timer
- A privacy pass that checks every response shape for hidden fields

8. Open issues

Four decisions are still open; each has a working default in this SRS until it is settled.

- [ ] Can a review's author delete their own review? Default: no, only the superadmin can (FR-REV-05, FR-REV-06).
- [ ] Final product name. Default: StudyCircle.
- [ ] AI provider. Default: any provider behind `aiService` (NFR-MNT-02).
- [ ] Final limits: groups per user (20), groups created (5), review eligibility (3 days), AI requests (20 per day), re-request wait (7 days).

Appendix A. Known limitations

Limitation	Reason	Production fix
Calls capped at 5 people	WebRTC mesh: each browser uploads one stream per participant	Move to an SFU such as LiveKit or mediasoup
Calls may fail on strict college or office networks	No TURN relay server	Add a TURN server (coturn or a hosted provider)
Mobile numbers not verified	SMS OTP costs money	Add OTP through an SMS provider (FR-AUTH-11)
AI answers can be wrong	Language models can make mistakes	AI label on every answer (FR-AI-06); RAG with note citations (FR-AI-10)